

# Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

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# The invisible layer

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Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

**92%** of researchers use software; **69%** say their research would not be practical without it.

The people keeping it running are increasingly called

**Research Software Engineers (RSEs)**, a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.

Hettrick et al. (SSJ, 2014/2022) ·  
Hoffmann, Nagle & Zhou (2024)  
doi:10.2139/ssrn.4693148



Jupyter



Python



R



Git



Spark



Pandas



NumPy



Docker



Ray



ggplot2



dplyr



Snakemake



Nextflow

# The UC research-to-infrastructure pipeline

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Data professionals built open data infrastructure: FAIR, DMPs, curation, provenance. **FAIR4RS** is the same inflection point for software — and UC has been showing the way for 50 years.

BSD Unix · Ceph · Apache Spark · Jupyter · RISC-V · Ray

The pipeline from university research to global infrastructure is not rare or accidental.

Barker et al. (2022)  
doi:10.1038/s41597-022-01710-x

UC Berkeley |



Jupyter



Apache Spark



RISC-V



Ray



BSD Unix

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**UC Santa Cruz** |



Ceph

**UCSC Genome Browser**

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**UC San Diego** |

**Cytoscape**

**EEGLAB**



UCSD Pascal

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**UCLA** |



Processing



Named Data Networking

**tz database**

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**UC San Francisco** |

**ChimeraX**



OpenMM

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**UC Davis** |



sourmash

**ERPLAB**

# What is an academic OSPO?

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## An Open Source Program Office

supports, governs, and promotes open source within an institution.

- Originated in industry:  
~77% of large tech companies have one
- First academic OSPO:  
Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



● UC OSPO Network · ● CURIOSS member

World map showing 34 CURIOSS member institutions. Six U

# The UC OSPO Network

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A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
  - Launched April 2024, funded by the Alfred P. Sloan Foundation
  - **\$1.85M** in external funding to date
  - Lead: UC Santa Cruz, the first OSPO in a large state university system
  - Serves 280,000+ students and 25,000+ faculty
- Acts as a **neutral convener**: no single campus, company, or grant

cycle can pull the work  
away.

Ruff (2026) UC Open  
Summit .

[youtu.be/eBriL3CDNeo](https://youtu.be/eBriL3CDNeo)



Map of California showing six UC OSPO Network campuses:

# Discovery · Sustainability · Education

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## Discovery

Mapping who  
does what  
across the UC  
system

- 236,000+  
repos  
scanned;  
~82,000  
institutionally  
affiliated
- 294-  
respondent  
system-wide  
survey
- UC Open  
Repository  
Browser (UC  
ORB)



## **Sustainability**

Keeping  
projects and  
communities  
healthy

- **58%** of UC contributors are also maintainers
- Cross-campus licensing working group (UCOP + tech transfer)
- Project health assessment (OSSPREY)



## **Education**

Coordinated  
training across

campuses

- Gap analysis  
→ curriculum  
inventory →  
learning  
pathways
- Published at  
**ucospo.net/education**
- Coordinated  
with  
Carpentries  
infrastructure

Gomez et al. (2025) arxiv:2506.18359 · Scarlett et al. (2025) doi:10.31235/osf.io/p8bx6\_v1

## Lessons from two years

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What the network model enables that single campuses cannot:

- .. **Shared staffing:** community manager, licensing specialist, technical roles
- 2. **Policy leverage:** the network has standing with UCOP; individual campuses do not
- 3. **Data at scale:** the GitHub and survey analyses require system-wide scope to mean anything
- 4. **Equity:** smaller campuses get services they could not build alone
- 5. **A replicable model:** \$1.85M produced something other state systems can follow

Each campus contributes unique expertise; the network routes it to where it's needed.

## Where this meets your work

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For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

### **For your institution:**

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

# You're already doing this work

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If you do technical data services, researchers are already bringing you these questions:

- “What license should I use for this code?”
- “Is this library still maintained? Should I depend on it?”
- “How do I make this reproducible and citable?”
- “My funder wants a software management plan.”
- “How do I accept contributions from collaborators?”

**That is OSPO work.** The frame makes it legible to your institution.

**The network has already built the resources** (Laura Langdon, UC OSPO):

[README](#) · [LICENSE](#) · [CONTRIBUTING](#) ·

**CODE OF CONDUCT** ·**CITATION.cff** + Zenodo DOI**· Growing a Community ·**  
**Carpentries Incubator**  
**lesson****Raise the floor, not  
just solve the ticket**

Fixing their immediate  
problem gets them  
through the week.

Moving them from “it  
runs on my machine” to  
maintainable, licensed,  
citable software gets  
their whole lab there,  
and their students after  
them.

You don’t need a formal  
OSPO to do this. You  
need the frame.

**OSPO heuristics  
already in your toolkit:**

health signals · bus  
factor · governance files  
· FAIR4RS · software  
DMPs · contributor

conventions · community  
governance

## Learn more / connect

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**UC OSPO Network: [ucospo.net](https://ucospo.net)**

- Education site: [ucospo.net/education](https://ucospo.net/education)
- Mailing list, Slack, events calendar

**Global network: [curiOSS.org](https://curiOSS.org), 34 academic OSPOs worldwide and growing**

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Slides: [tinyurl.com/iassist2026](https://tinyurl.com/iassist2026)



Handout:

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Speaker notes